

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Secure and Automated Credit Card Processing System

Course: Software Engineering and Agile Development Lab – Exercise 1

Version: 1.0

Date: 6/11/2026

1.Team Members & Roles

- Divya R – Project Manager
- Pudi Pavani – Lead Developer
- Shree Vaishnavi M – QA Lead
- Pratiksha L – Documentation Lead

2.Team Agreement (Working Norms)

- All members must attend weekly meetings.
- Work must be equally distributed.
- Deadlines must be strictly followed.
- Communication will be through WhatsApp and Google Meet.
- Respect and cooperation among team members is mandatory.
- Any conflict will be resolved by discussion.
- Plagiarism is strictly prohibited.
- Final documents will be reviewed by all members before submission.

3.Problem Statement Development

3.1 Project Title

Secure and Automated Credit Card Processing System

3.2 Problem

Existing credit card systems depend on manual and time-consuming processes for application, document verification, and approval. After issuance, lack of real-time transaction monitoring increases fraud risk and unauthorized transactions.

3.3 Proposed Solution

Develop a secure automated system that integrates:

- Online credit card application

- Automated document verification
- Approval workflow
- Continuous transaction monitoring
- Fraud detection and alerts

3.4 Target Users

- Bank customers
- Bank employees
- Fraud analysts
- System administrators

3.5 Value Proposition

- Faster processing
- Improved security
- Reduced fraud
- Better user experience

4. Stakeholder Analysis

4.1 Primary Stakeholders

- Customers applying for credit cards
- Bank employees responsible for verification and approval

4.2 Secondary Stakeholders

- Bank managers
- Fraud detection team
- System administrator

5. Stakeholder Interests & Influence

Stakeholder	Interest	Influence
Customer	Fast approval and security	Medium
Bank Employee	Easy verification	High
Manager	Reports and fraud alerts	High
Fraud Team	Detect fraud	High
System Admin	System maintenance	Medium

6. Stakeholder Register

ID	Stakeholder	Role	Interest	Influence
S1	Customer	End user	Apply for card	Medium
S2	Bank Employee	Verifier	Process applications	High
S3	Manager	Decision maker	View reports	High
S4	Fraud Analyst	Security	Detect fraud	High
S5	Admin	Maintenance	Manage system	Medium

7. Scope Definition – In-Scope

- User registration and login
- Online credit card application
- Document upload
- Application verification and approval
- User notifications
- Transaction tracking
- Fraud detection
- Report generation

8. Scope Definition – Out-of-Scope

- Physical card printing
- Real payment gateway integration
- Mobile application
- Biometric authentication
- Cryptocurrency transactions
- International banking systems

9. Scope Boundaries

Included

- Application submission
- Verification
- Approval
- Monitoring
- Fraud alerts

Excluded

- Core banking integration
- Hardware installation
- Physical card delivery

10. Introduction

10.1 Purpose

This document specifies the requirements of the Secure and Automated Credit Card Processing System. It serves as a reference for developers, testers, and instructors.

10.2 Scope

The system automates credit card application processing, verification, approval, and fraud monitoring to reduce delays and improve security.

10.3 Definitions, Acronyms

- SRS – Software Requirements Specification
- OTP – One Time Password
- KYC – Know Your Customer
- FR – Functional Requirement
- NFR – Non-Functional Requirement

11. Overall Description

11.1 Product Perspective

The system is a web-based application that replaces manual credit card processing.

11.2 Product Functions

- User registration
- Application submission
- Document verification
- Approval workflow
- Transaction monitoring
- Fraud detection
- Report generation

11.3 User Characteristics

- Customer – basic computer knowledge

- Bank employee – trained user
- Admin – technical knowledge

11.4 Constraints

- No real banking transactions
- No real payment gateway
- Academic project only
- Internet required

12. Functional Requirements

FR-01: User Registration

System shall allow users to register.

Acceptance: valid email, password ≥ 8 characters

FR-02: User Login

System shall allow secure login.

Acceptance: valid credentials redirect to dashboard

FR-03: Application Submission

System shall allow users to apply for credit card.

Acceptance: form saved and confirmation shown

FR-04: Document Upload

System shall allow upload of ID and income proof.

Acceptance: valid file formats only

FR-05: Application Verification

Admin shall verify applications.

Acceptance: approve or reject

FR-06: User Notification

System shall notify user of status.

Acceptance: visible on dashboard

FR-07: Transaction Monitoring

System shall log all transactions.

Acceptance: amount and time stored

FR-08: Fraud Detection

System shall detect abnormal transactions.

Acceptance: suspicious transactions flagged

FR-09: Report Generation

System shall generate reports.

Acceptance: transaction and fraud reports

13. Non-Functional Requirements

NFR-01 Performance:

System should load pages within 3 seconds.

NFR-02 Security:

Passwords and sensitive data must be encrypted.

NFR-03 Usability:

System must be easy to use.

NFR-04 Reliability:

System uptime must be at least 95%.

14. Use Case Diagram (Appendix)

Actors

- Customer
- Admin
- Manager

Use Cases

- Register
- Login
- Apply for card
- Upload documents
- Approve application
- Monitor transactions
- Generate reports

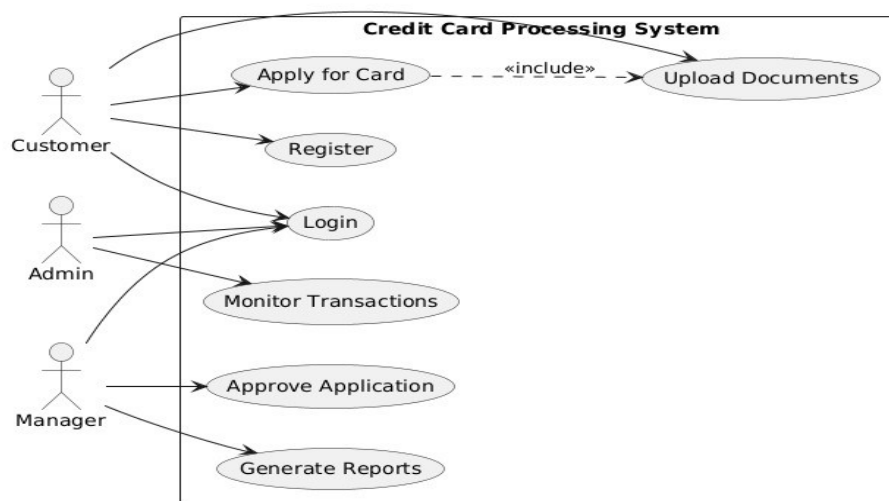


Figure 1: UML Use Case Diagram Representing the Functional Interactions in the Credit Card Processing System

15. Glossary

Term	Meaning
KYC	Identity verification
Fraud	Unauthorized transaction
OTP	One time password
Dashboard	Main control screen
Transaction	Payment made using card

16. Project Charter

Project Name: Secure and Automated Credit Card Processing System

Objective: Automate and secure credit card processing

Sponsor: Banking Organization

Duration: 3 months

Deliverables: SRS, diagrams, prototype

Budget: Academic project

17. Stakeholder Register Submission

(Refer Section 6 Stakeholder Register)

18. Complete SRS Document

This document contains:

- Introduction
- Overall Description
- Functional Requirements
- Non-Functional Requirements
- Use Case Diagram
- Glossary

19. Requirements Sign-Off

Instructor Name: _____

Signature: _____

Date: _____

Team Leader: _____

Signature: _____